NAMAN YESHWANTH KUMAR

linkedin.com/in/namany | github.com/naman-ranka | nyeshwan@asu.edu | (623) 249-1265

SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) building self-improving agents that close the loop from RTL spec to GDSII. SiliconCrew — an autonomous LangGraph agent that reads VCD waveforms, diagnoses signal-level bugs, patches RTL, and re-verifies without human input — scored **37/92 first-pass on CVDP** and **5/5 on ASU Spec2Tapeout**. Drove a GNN ASIC end-to-end through Synopsys Design Compiler synthesis, Cadence Innovus place-and-route, and Voltus power sign-off on ASAP7 7nm.

TECHNICAL SKILLS

AI for Chip Design: LangGraph | LangChain | ReAct Agents | MCP Protocol | Multi-provider LLM Routing | PyTorch | TensorFlow | CUDA | Python | FastAPI

Synthesis, PD & Sign-off: Synopsys Design Compiler | Cadence Innovus | Voltus | Calibre DRC/LVS | OpenROAD | Yosys | SymbiYosys | Static Timing Analysis | NLDM Liberty | SDC | Equivalence Checking

RTL & Architecture: SystemVerilog | Verilog | Cocotb | iverilog | ASAP7 7nm FinFET | SkyWater 130nm | gem5 | C++ | Docker | Linux

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0) — **May 2026**
Arizona State University — **Tempe, AZ**
Coursework: VLSI Design · VLSI Design Automation · Advanced Computer Architecture · Algorithm/Hardware Co-Design · SoC Design

Bachelor of Engineering — Electrical and Electronics — **August 2022**
BMS College of Engineering — **Bengaluru, India**

PROJECT EXPERIENCE

SiliconCrew — Self-Improving Agent for RTL-to-GDSII Chip Design — **Nov 2025 – Present**

- Built SiliconCrew, an autonomous LangGraph ReAct agent (21 tools) that drives the complete chip design pipeline — RTL generation → lint → simulation → Yosys synthesis → OpenROAD place-and-route → post-synthesis simulation → GDSII — across SkyWater 130nm and ASAP7 7nm PDKs, with SymbiYosys bounded model checking and Cocotb constrained-random verification; FastAPI + WebSocket backend, full MCP server (stdio/SSE/HTTP) compatible with Claude Desktop, VS Code, and any MCP client.
- Closes the loop without human intervention: on simulation failure the agent reads VCD waveforms, diagnoses signal-level bugs, patches RTL, and re-verifies — system prompt mandates "never guess, always read waveforms" and enforces a minimum of 3 improvement iterations until all goals are met; enforces SDC constraint validation, Yosys RTL-vs-netlist equivalence checking, and signoff guardrails before tape-out.
- Achieved **37/92 first-pass on CVDP** (Comprehensive Verilog Design Problems) and **5/5 on ASU Spec2Tapeout** with zero human intervention; multi-provider LLM routing (Gemini, OpenAI, Anthropic) with provider-specific quirk handling and per-session token and cost tracking.

Graph Neural Network ASIC Accelerator — Full RTL-to-GDSII Sign-off on ASAP7 7nm — **Spring 2025**

- Designed an 11-module GNN ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and drove the design through industry sign-off flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus place-and-route with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved **50,300 µm²** chip area at 50.6% core density, **52.29 mW** total power (64.8% switching), and **99.2 ns** end-to-end latency.

Static Timing Analysis Engine — Python (Sign-off-grade) — **Fall 2025 – Spring 2026**

- Built a complete Static Timing Analysis engine in Python from scratch — Liberty NLDM parser (7×7 delay/slew LUTs with bilinear interpolation), .bench circuit-graph parser, topological sort via Kahn's algorithm, forward arrival-time and slew propagation, backward RAT/slack computation, and critical path extraction; validated against ISCAS benchmarks c17, b15, and c7552.

5-Stage Pipelined MIPS Processor — SystemVerilog — **Fall 2025 – Spring 2026**

- Implemented a 5-stage pipelined MIPS processor in SystemVerilog (33 instructions) with a 12-path forwarding network (EX-to-EX, MEM-to-EX, WB-to-ID, branch, JR) and 6 hazard stall conditions; extended with an 8-entry fully-associative CAM instruction cache (3-state FIFO FSM) and LL/SC atomic operations with link-address tracking.

Trezzit — Production AI System, Shipped Solo (trezzit.com) — **Feb 2025 – Present**

- Sole-built and shipped a production AI platform serving **600+ users** for 12+ months — multi-provider inference pipeline (Gemini, OpenAI, Claude, Grok) iterated from **~60% → 95%+** accuracy via two-pass semantic validation and automatic model-upgrade fallback; **995 automated tests**, GitHub Actions CI/CD, full prod/staging infra on DigitalOcean + Vercel + Supabase.

WORK EXPERIENCE

Systems Engineer — **Oct 2022 – Apr 2024**
NCR GOLD — **Bengaluru, India**

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

Research Intern — Propulsion & Power Electronics — **Oct 2021 – Aug 2022**
Indian Institute of Science — **Bengaluru, India**

- Applied Ansys Maxwell FEA to optimize a 1.5 kW axial flux BLDC motor — redesigned stator geometry and upgraded magnetic materials to improve flux linkage and reduce core losses; modeled FOC control and battery management in Simulink for eVTOL propulsion reliability.

AWARDS & HONORS

- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.